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20 - 30 % time
Revision

70 - 80 %
practice

→ Basic Revision

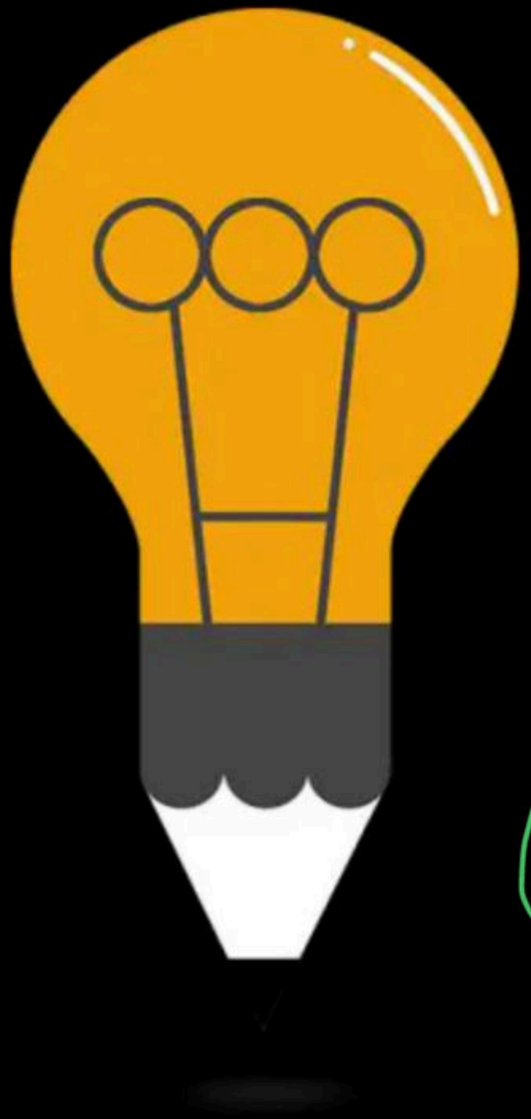


Basics and CPU Registers

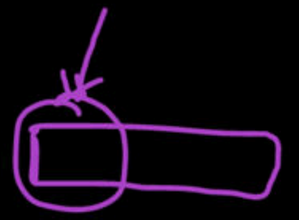
Revision & Practice Course on COA

Vishvadeep Gothi • Lesson 1 • Oct 31, 2023

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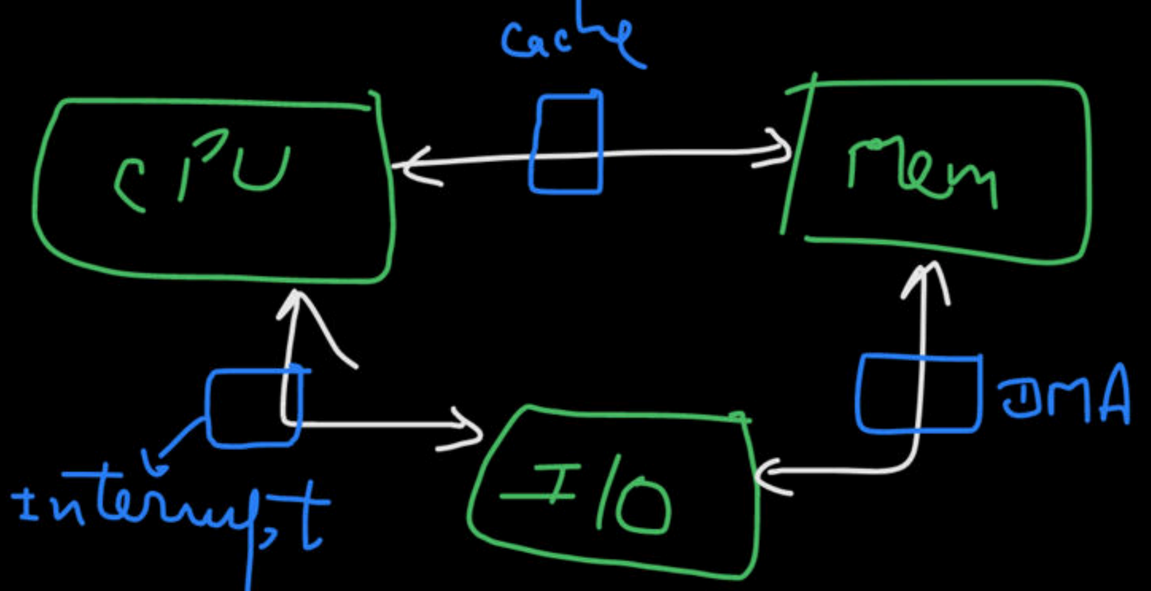


Co CA → CPU design
Instⁿ
Add. modes
Data format



RIB COA: 1

By: Vishvadeep Gothi



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Question

div . . .

Consider the following statements :

1. A computer will have a divide instruction ← arch.

2. Divide instruction will be implemented by a special division unit ← org.

Which of the following is correct?

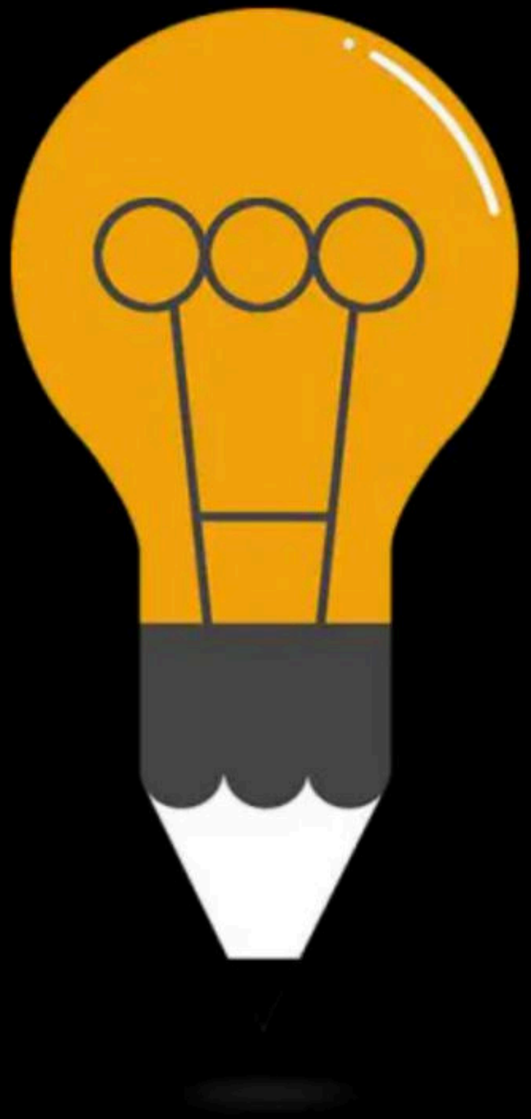
(a) Both 1 and 2 are not architectural design issues.

(b) Both 1 and 2 are not organizational issues.

(c) 1 is an architectural design issue while
2 is an organizational issue.

(d) 1 is an organizational issue while
2 is an architectural design issue.

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Basics of Computer Systems

By: Vishvadeep Gothi

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Prerequisites

- Basic components of computer: CPU, memory (RAM, ROM, HDD), I/O
- Number system: Binary, Decimal, Hexadecimal etc.
- Digital logic basics: Mux, Decoder etc.

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Prerequisites

- Powers of 2:

Unit	Time	Bit or Byte
K (Kilo)	10^3	2^{10}
M (Mega)	10^6	2^{20}
G (Giga)	10^9	2^{30}
T (Tera)	10^{12}	2^{40}

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Why COA

- To understand: **How a computer works**
- To understand other courses: **OS, Compiler, Programming** etc.
- Help in real world development: **DBMS, Hardware Design, IoT problems** etc.

Computer Organization & Architecture

Computer Organization & Architecture

Computer Architecture:

Conceptual design and fundamental operational structure.

Computer Organization & Architecture

Computer Architecture:

Conceptual design and fundamental operational structure.

Computer Organization:

- Deals with physical devices and their interconnections
- With a perspective of improving the performance.

Computer Organization & Architecture

Computer Architecture	Computer Organization
• CPU Design	• I/O Organization
• Instructions	• Memory Organization
• Addressing modes	• Performance
• Data format	

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Data In Computers

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Binary Numbers

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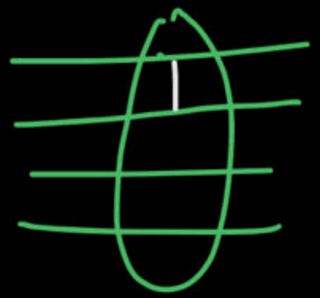
Components of Computer

- CPU
  - C.U.
  - A.L.U.
- Memory:
  - Main memory
  - Sec. memory
- I/O Devices:
  - Input
  - output
  - storage

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Other Components

- System Buses
- CPU Registers



⇒ 16-bits wide bus

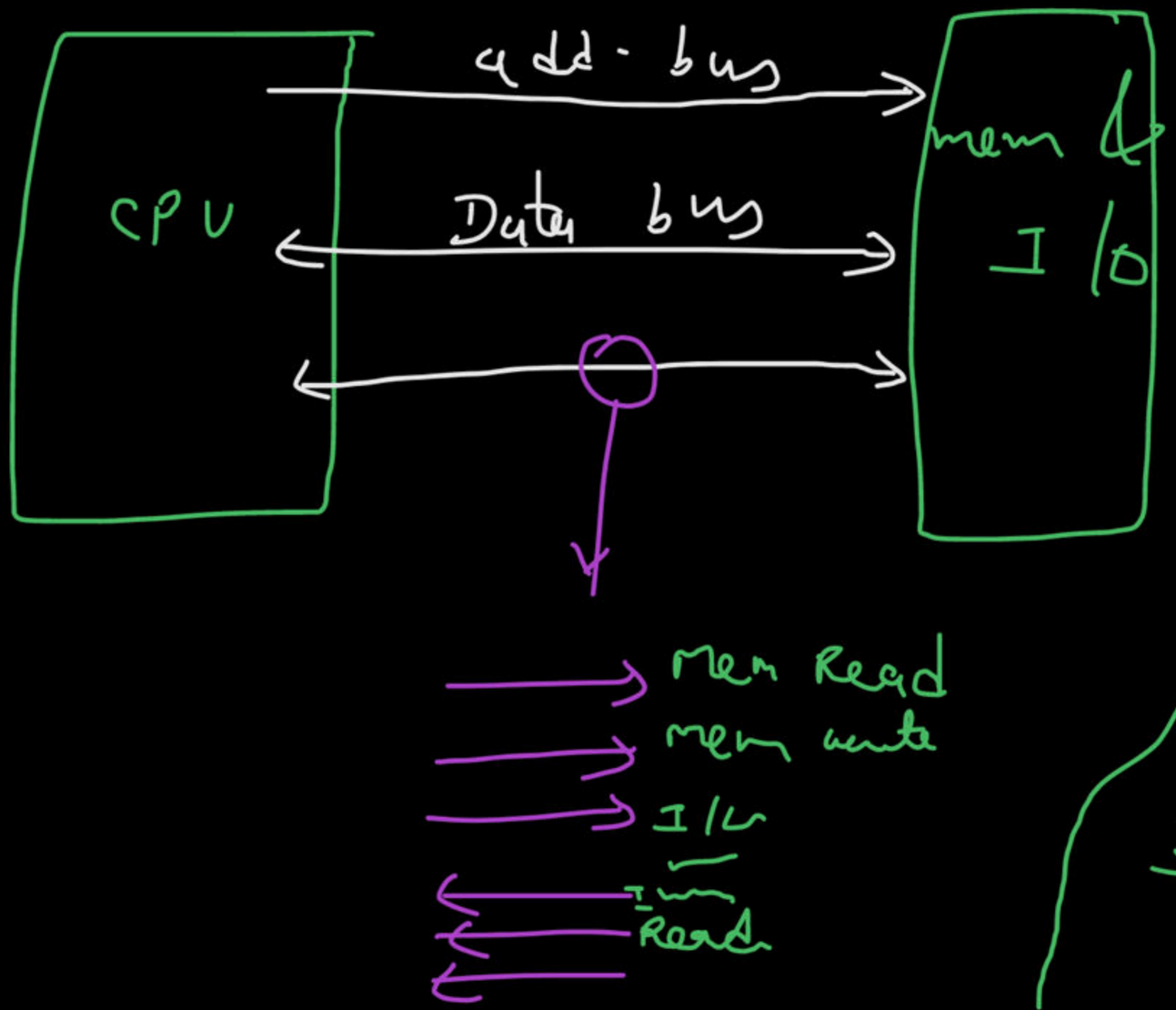
⇒ 16-lines bus

Other Components

- System Buses:
  - Address Bus → add. from CPU to mem & I/O
  - Data Bus ⟷ data
  - Control Bus → control signals

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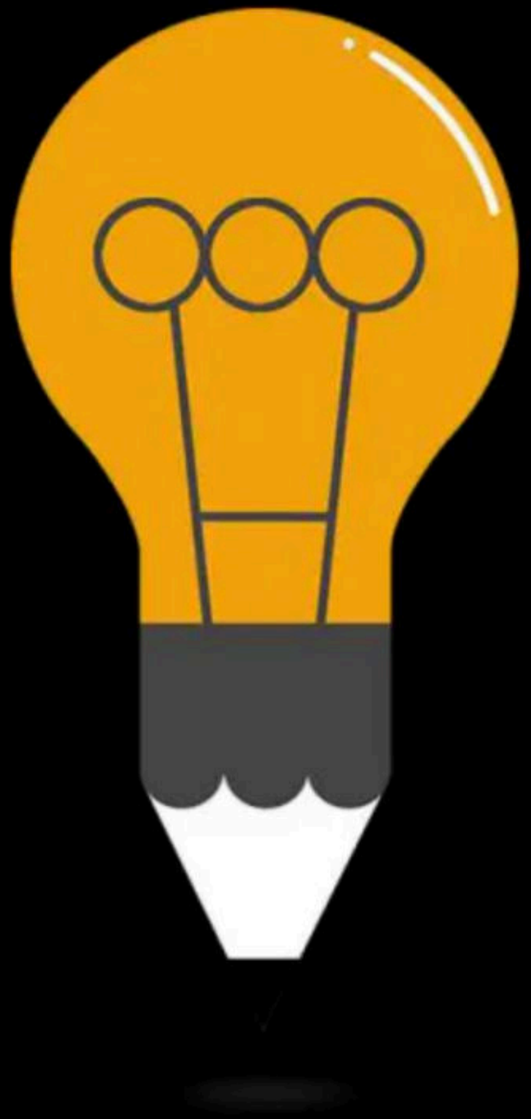
System Buses



Data bus content :

1) operand of any type
2) Address
   - for indirect mode
   - for storing addr on stack
   - for restoring PC from stack
3) Instⁿ

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CPU Registers

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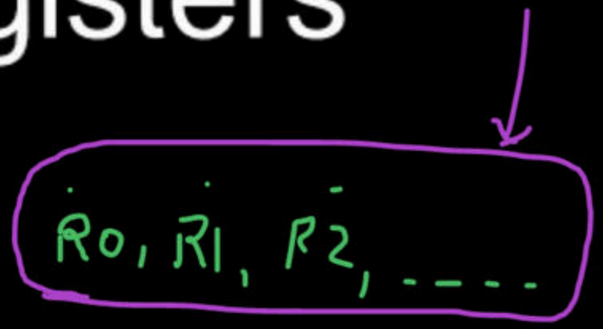
Motivation

Hard-work can beat the talent

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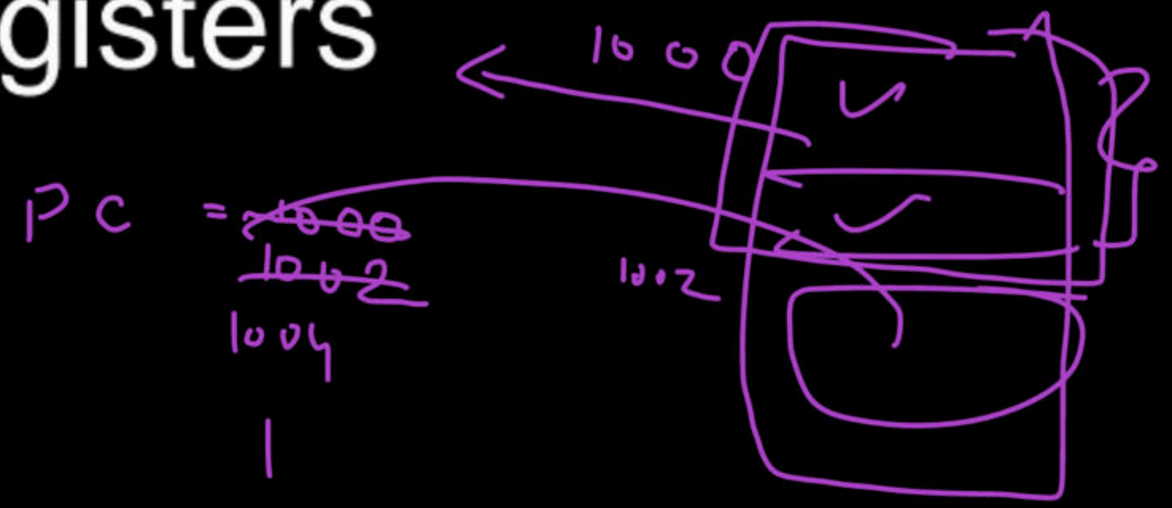
CPU Registers

- CPU Register
  - General Purpose Registers (GPRs)
  - Special Purpose Registers



32-bits
64-bits
16-bits
8-bits

CPU Registers

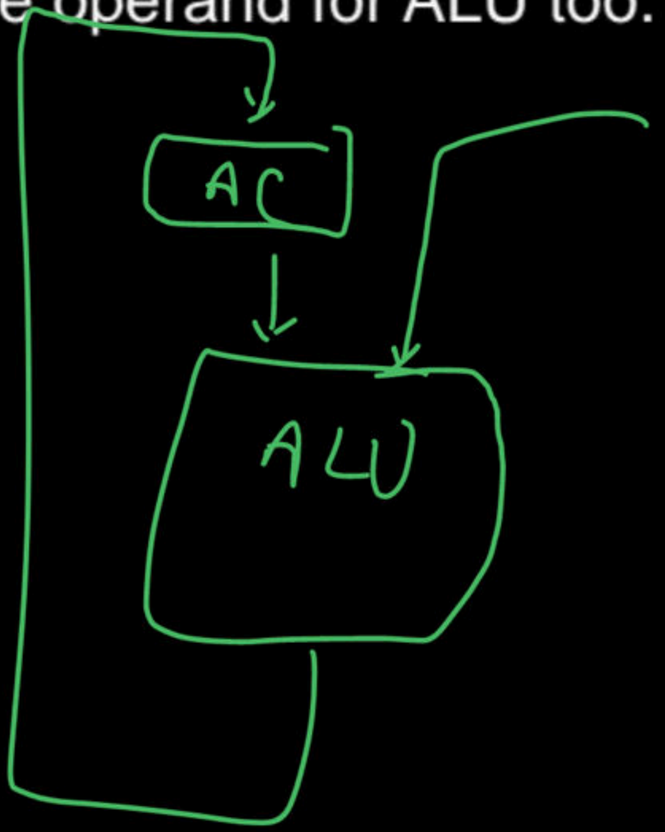


- CPU Register
  - General Purpose Registers (GPRs)
  - Special Purpose Registers
    1. Accumulator (AC)
    2. Program Counter (PC)
    3. Instruction Register (IR)
    4. Stack Pointer (SP)
    5. Flag Register / Program Status Word (PSW)
    6. Address Register (AR) / Memory Address Register (MAR)
    7. Data Register (DR) / Memory Data Register (MDR) / MBR

Accumulator

Used to store result of ALU and sometimes on of the operand for ALU too.





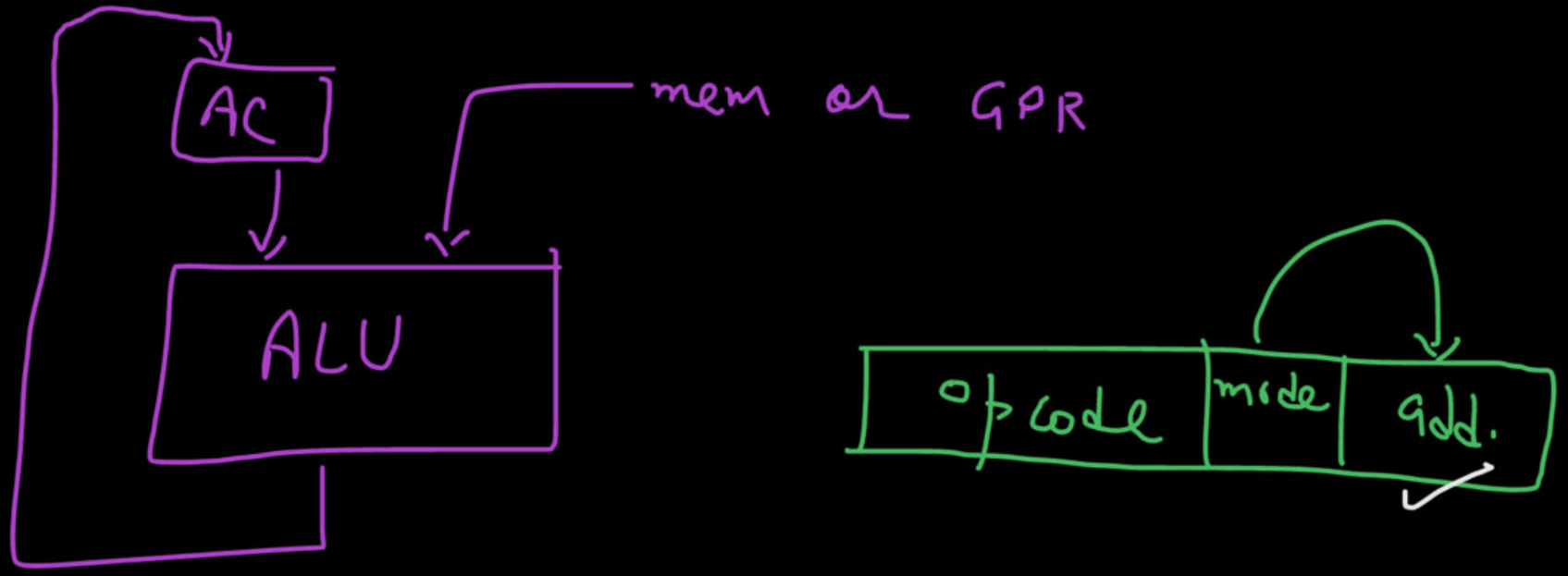
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Types of Architecture

Based on ALU input:

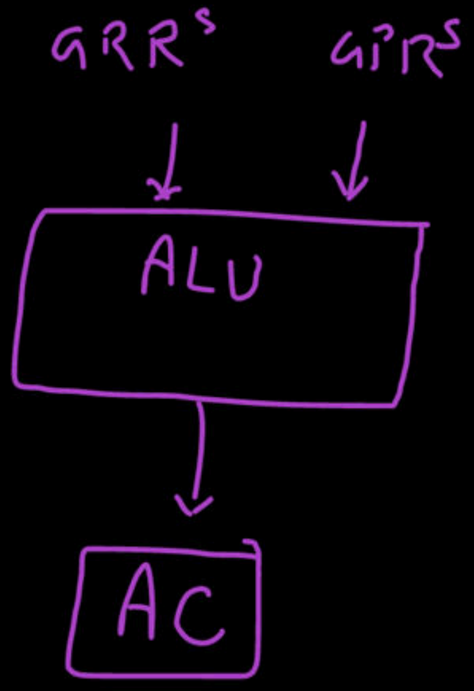
1. AC-Based Architecture
2. Register Based Architecture
3. Register-Memory Based Architecture
4. Complex System Architecture
5. Stack Based Architecture

AC-Based Architecture



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Register-Based Architecture



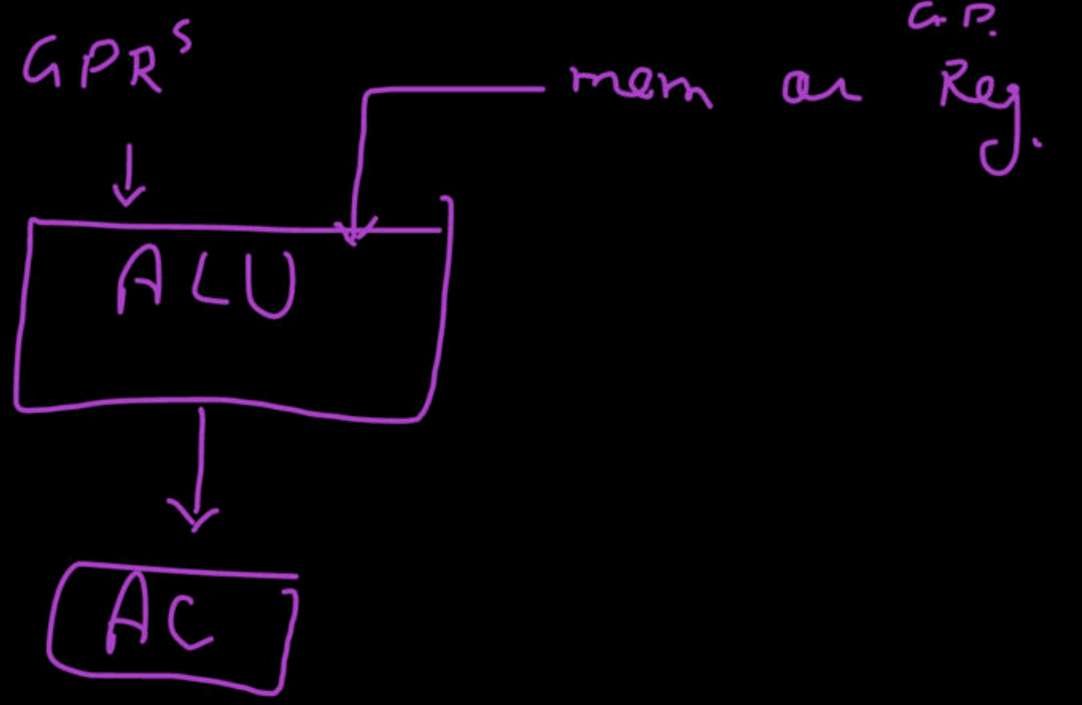
a+b

R1 ← a

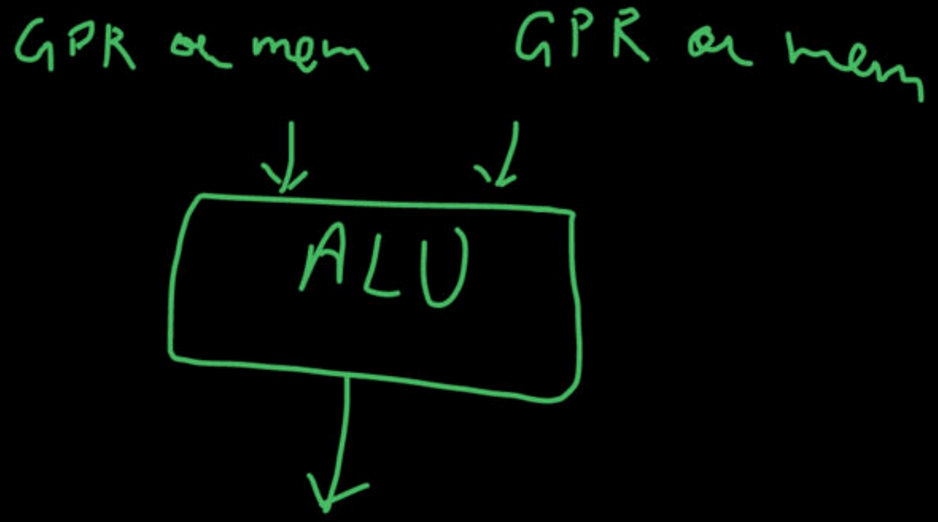
R2 ← b

← R1 + R2

Register-Memory Based Architecture

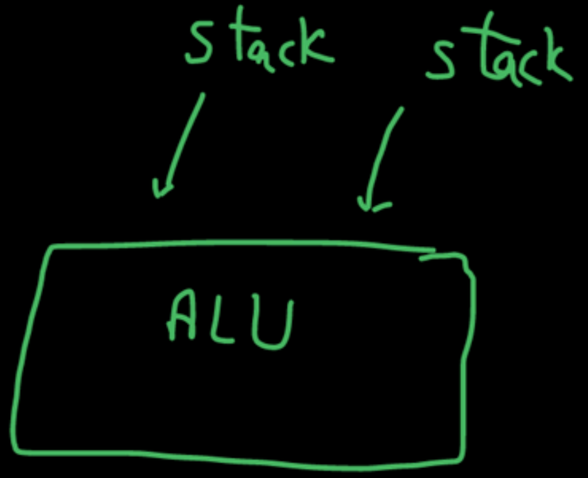


Complex System Architecture



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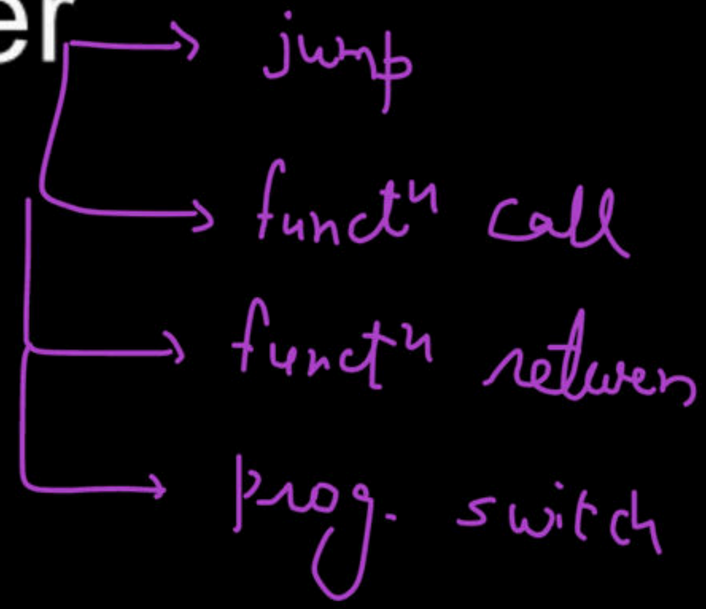
Stack-Based Architecture



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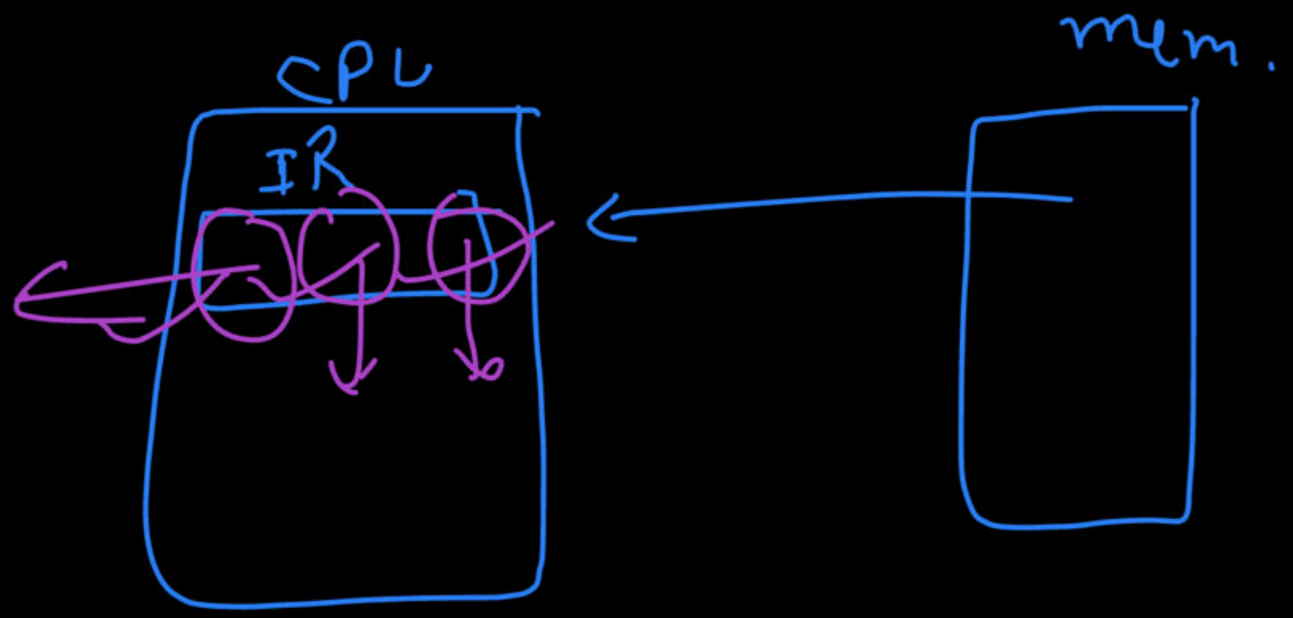
Program Counter

Stores address of next instruction to be executed



Instruction Register

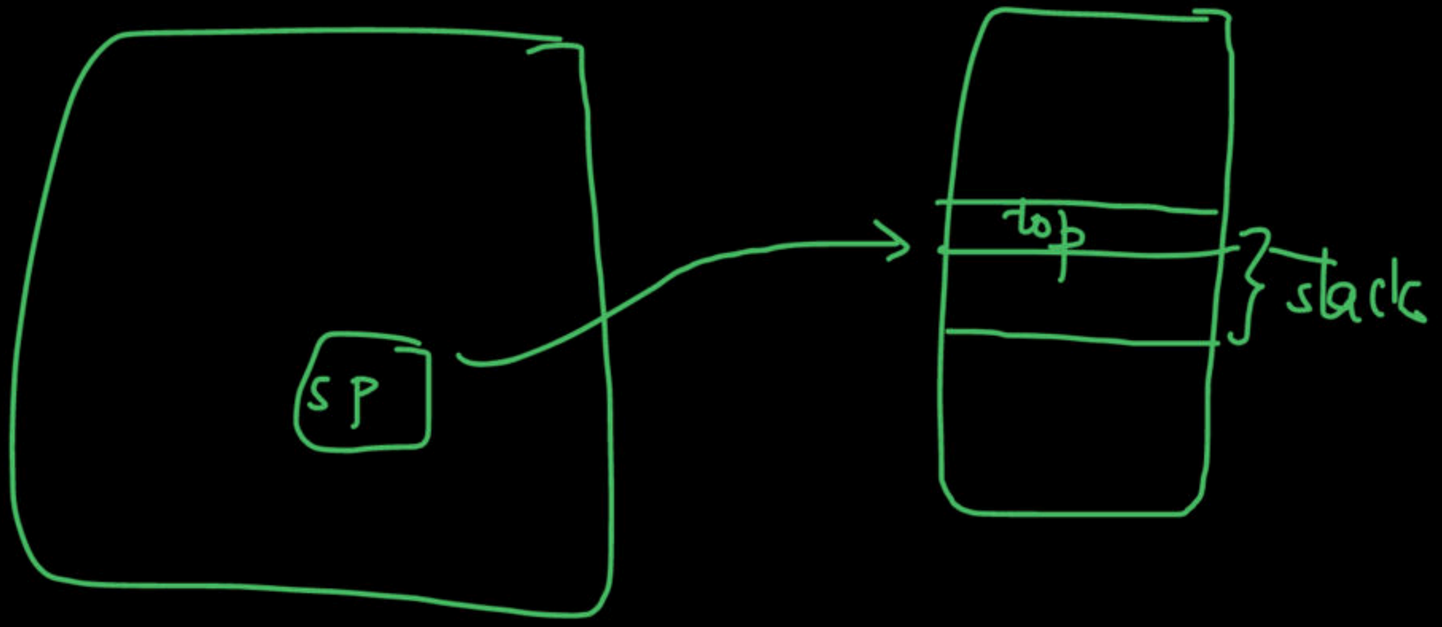
Stores the current instruction to be executed



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Stack Pointer

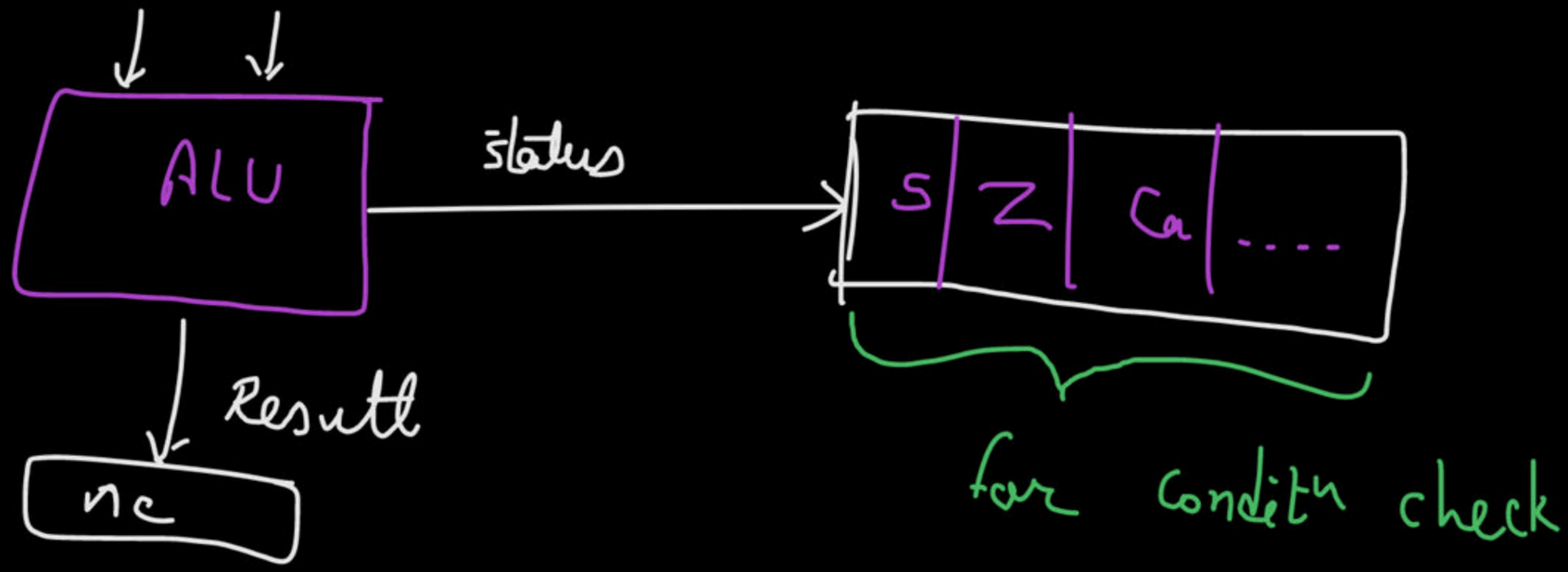
Stores the address of the top of the stack



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Flag or Status Register / PSW

Stores the status of the ALU result



Address Register or MAR

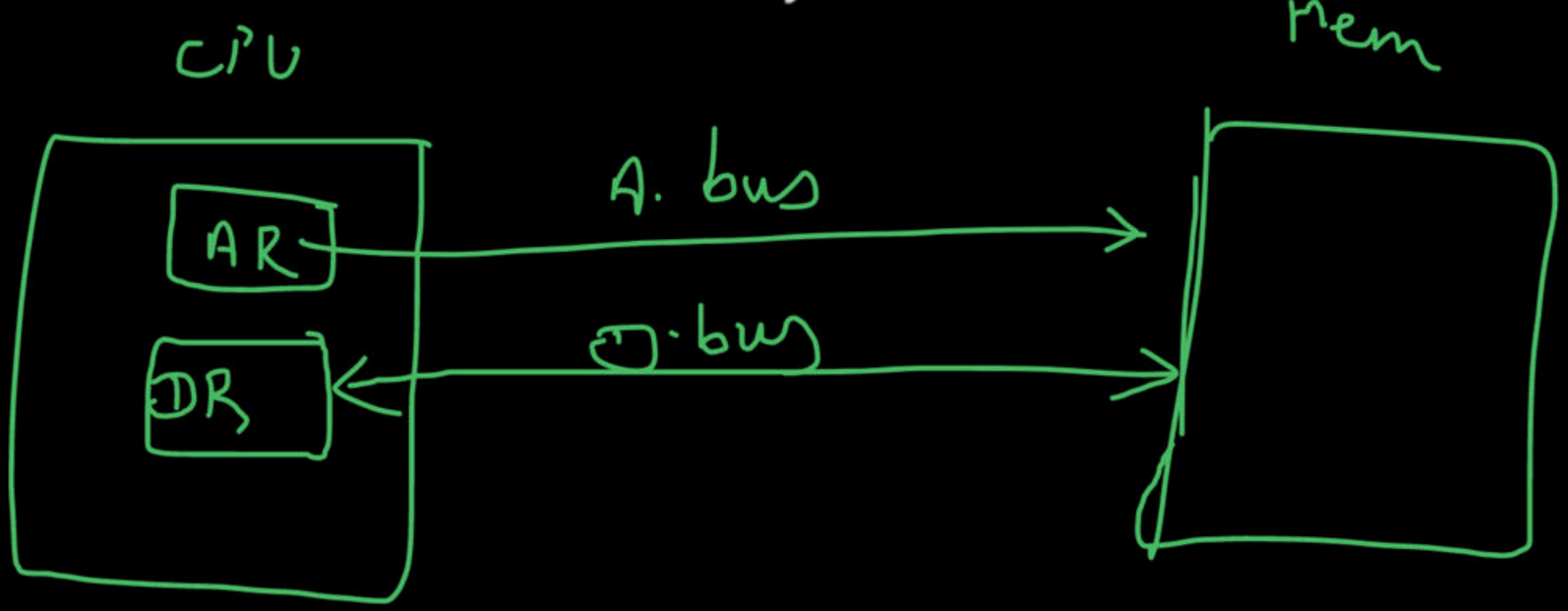
Used to send address to memory

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Data Register or MDR or MBR

Used to send data to memory

And to receive data from memory



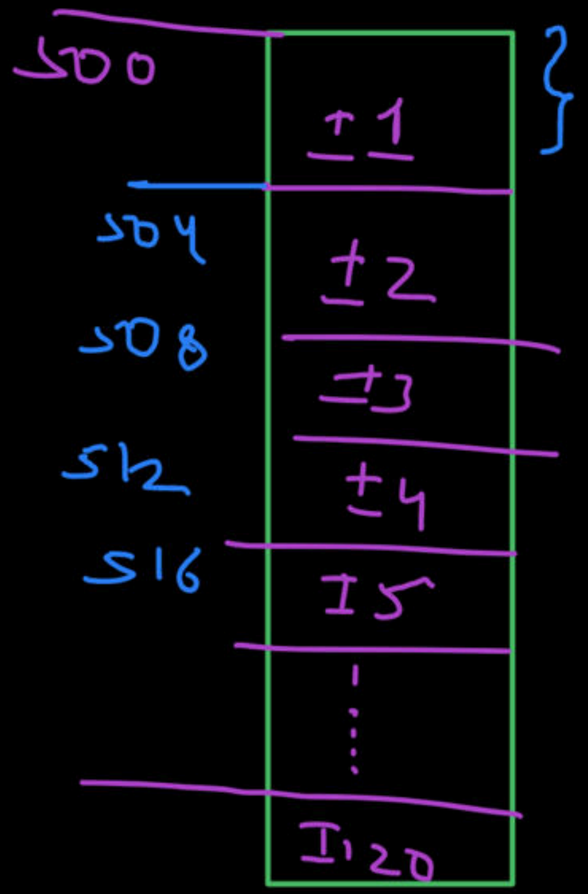
Question

A CPU has 4 bytes instructions. A program (Instructions I1 to I200) starts at address 500 (in decimal). Find the address of following instructions:

1. I1 = 500
2. I5 = 516
3. I120 = 500 + (119 * 4)
   = 976

4. add. of instⁿ i

= 500 + 4 * (i - 1)



when instⁿ I18 is in execⁿ then value of PC will be = ?

Ans:- add. of I19

= 500 + 4 * (19 - 1) =
572

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Question

A CPU has 4 bytes instructions. A program (Instructions I1 to I200) starts at address 500 (in decimal). What should be the PC value when instruction I6 will be executing in CPU?

= add. of I7

= 500 + 4 * (7-1)

= 524

Question

A CPU has 4 bytes instructions. A program (Instructions I1 to I200) starts at address 500 (in decimal). What should be the PC value when instruction i will be executing in CPU?

$= \text{add. of } i^{th} \text{ inst}^{n} \; i+1$

$= 500 + 4 * (i + 1 - 1)$

$= 500 + 4i$

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Question

3 bytes

A CPU has 24-bits instruction. A program starts at address 600 (in decimal). Which of the following is a legal program counter value?

(A) 700

(B) 800

(C) 900

(D) 950

PC = ~~600~~
~~603~~
~~606~~
609
⋮

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Question MSQ

Consider 5 instructions of a program are as follows:

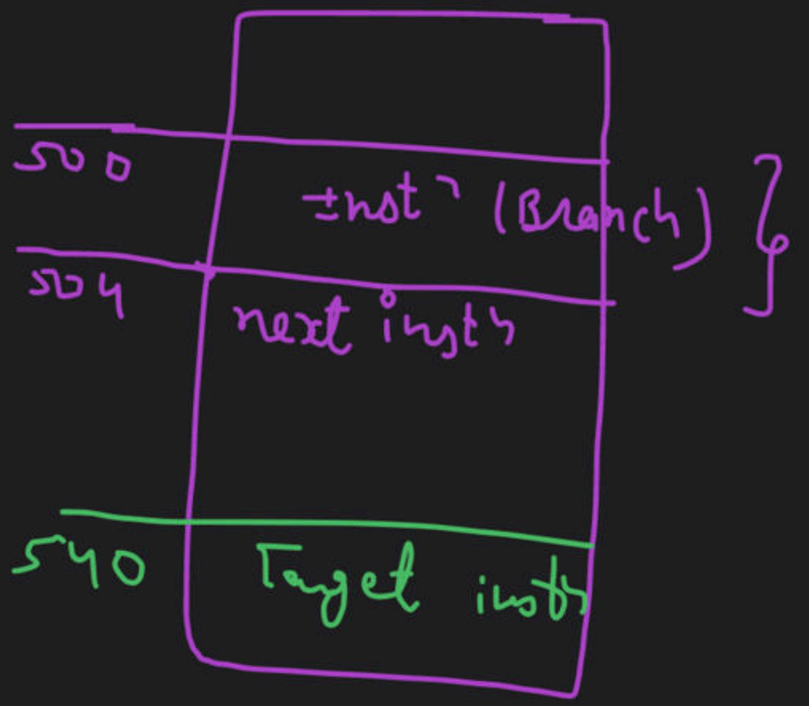
Instruction	Size in words
i	2
i+1	1
i+2	2
i+3	3
i+4	2

If the program is loaded from location 5000, and the memory is word addressable then value of PC immediately after the execution of instruction i+3 can be? (Note: All the numbers in decimal)

(A) 5000
(B) 5002
(C) 5004
(D) 5008

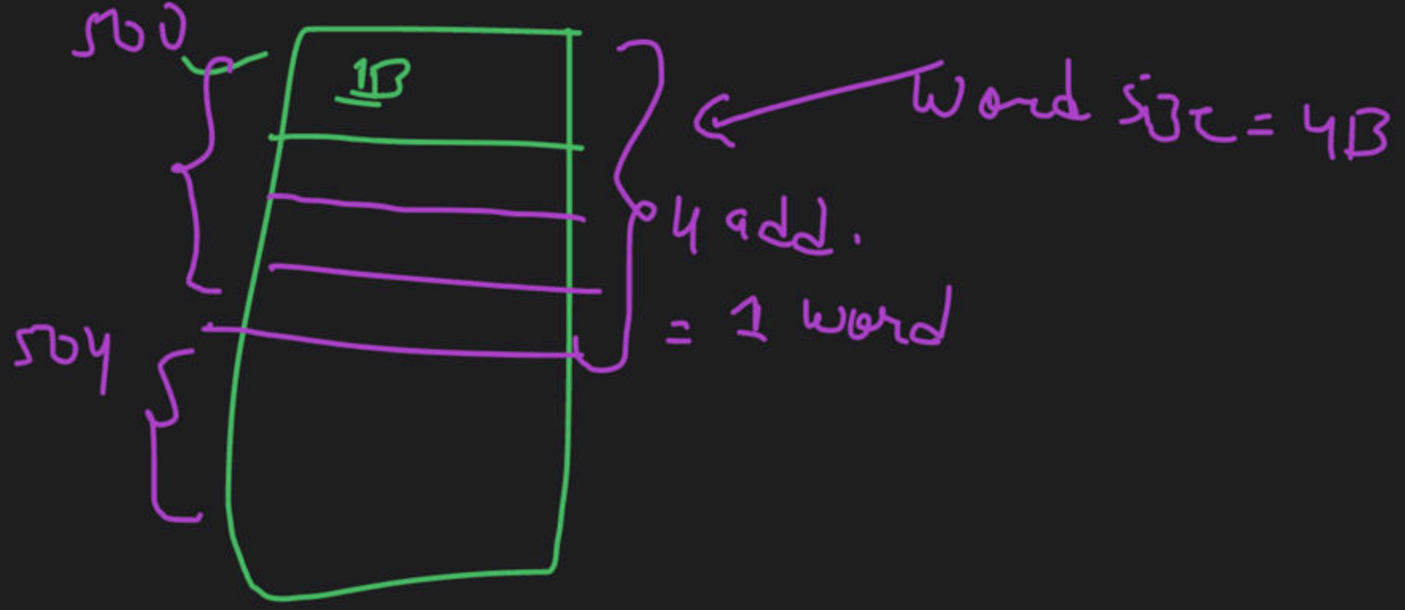
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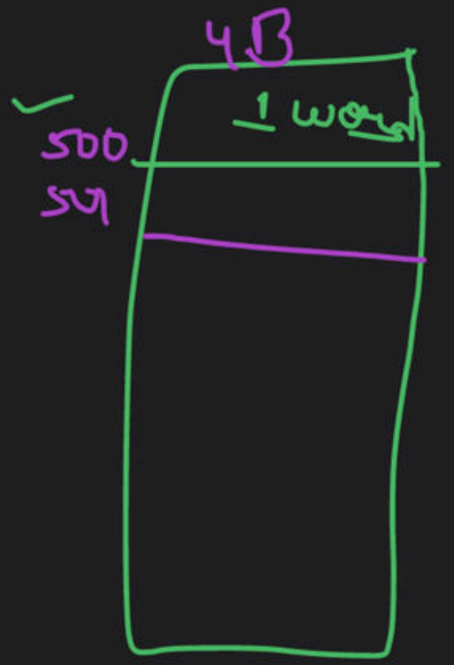
	Branch
IF	✓ PC = ~~500~~ 504
Decode	✓ Branch inst^n
Eff. add. calc.	Target add. = 540
Oper. fetch	×
Execution	Condition check & PC update PC = 540
W.B.	×



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Byte add. & word add.





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Question

Consider the following program segment. Here R1, R2 and R3 are the general purpose register. Assume that the content of memory location 3000 is 50 and location 2000 is 25. Content of register R2 is 12. All numbers are in decimal. After the execution of this program the value of memory location 2000 is?

Instructions	Operations
MOV R1, #15	R1 ← #15
MOV (2000), R1	M[2000] ← R1
ADD R2, (2000)	R2 ← R2 + M[2000]
MOV(3000), R2	M[3000] ← R2
MOV R3, R1	R3 ← R1
ADD R3, (3000)	R3 ← R3 + M[3000]
MOV (2000), R3	M[2000]← R3
HALT	Stop

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Question

Consider the following program segment. Here R1 and R2 are the general-purpose register. Assume that the content of memory location 3000 is 14 and location 2000 is 23. Content of register R2 is 10. All numbers are in decimal. After the execution of this program the value of R2 is?

	Instructions	Operations
	MOV R1, #17	R1 ← #17
X:	DEC R1	R1 ← R1 – 1
	JNZ Y	Jump to Y on Non-Zero
	ADD R2, (3000)	R2 ← R2 + M[3000]
	MUL R2, 3	R2 ← R2 * 3
	JMP Z	Jump to Z
Y:	ADD R2, (2000)	R2 ← R2 + M[2000]
	JMP X	Jump to X
Z:	HALT	Stop

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Question

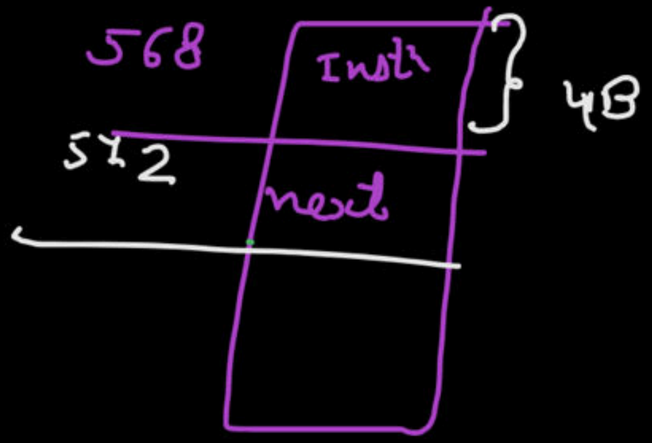
4B inst^n

Consider a computer which has 2-word instructions. 1 word size is 2 bytes. In main memory an instruction is stored at location 568 (decimal). The decimal value of PC f or byte addressable and word addressable memory?

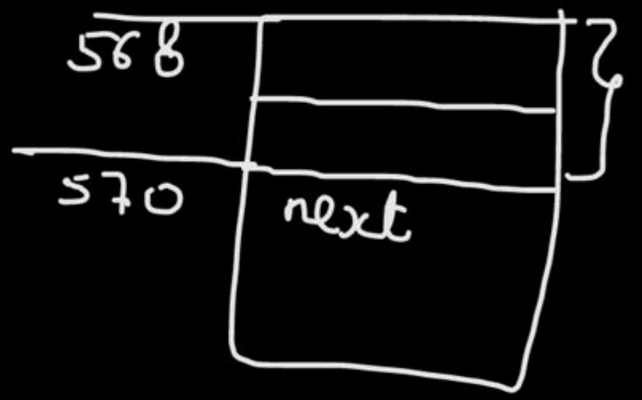
word = 2B

1. Before this instruction fetch 568 | 568
2. After this instruction fetch 572 | 570
3. After execution 572 or Target add. | 570 or Target add.

byte add.



word add.



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Question

Consider a system which support only 3 address instructions only, and supports 256B memory. If the instruction size is 40-bits then maximum & minimum number of instruction supported by the system are?

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Question

Consider a system which support only 2 address instructions only, and supports word addressable memory. The memory has total capacity of 2MB with word size of 4 Bytes. The system supports 350 distinct instructions. How many memory locations are required to store each instruction in the memory?

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Question

Consider a system which support only 2 address instructions only, and supports word addressable memory. The memory has total capacity of 2MB with word size of 4 Bytes. The system supports 350 distinct instructions. How many memory locations are required to store each instruction in the memory?

In the above question if a program is to be stored in the memory with 500 instructions, then the amount of memory required to store the entire program is _____ bytes?

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Question

The word addressable memory of a computer has 256K words of 32-bit each. The computer has an instruction format with four fields; an operation code field, a mode field to specify one of 8 addressing modes, a register address field to specify one of the 64 processor registers and a memory address field.

1. Specify the instruction format and the number of bits in each field if the instruction is stored exactly in one word in memory?
2. Maximum How many instructions supported by the computer?

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Question

The word addressable memory of a computer has 256K words of 32-bit each. The computer has an instruction format with four fields; an operation code field, a mode field to specify one of 8 addressing modes, a register address field to specify one of the 64 processor registers and a memory address field.

1. Specify the instruction format and the number of bits in each field if the instruction is stored exactly in one word in memory?
2. Maximum How many instructions supported by the computer?

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Question

A digital computer has a memory unit with 32-bits per word. The instruction set consists of 240 different operations. All the instructions have an operation code part (opcode) and an address part (allowed for only 1 address). Each instructions is stored in one word of memory.

1.How many bits are needed for opcode?

2.How many bits are left for address part of instruction?

3.What is the maximum allowable size of memory (word addressable)?

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Question

A computer supports only 3 address instructions with length 35-bits each. There are 129 distinct instructions supported by the system. If the memory used in the system is word addressable with word size of 32 bits. The maximum size memory supported by system is _____ KBytes?

Question

A computer system has x-instructions supported with 2^m KB memory size. The computer supports only 2-address instructions, the size of each instruction is _____ bits?

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Happy Learning.!

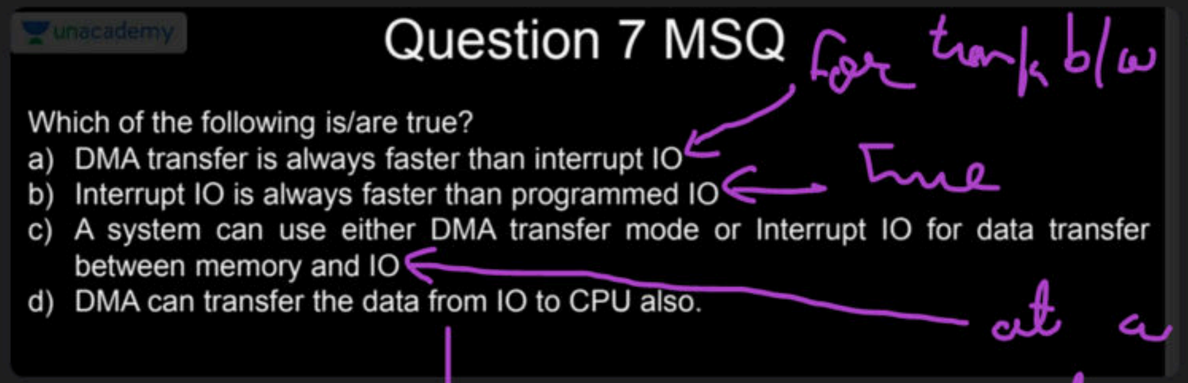




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▲ 1 • Asked by Diksha

ye wla quest bhi sir



for transfer b/w mem & I/O

True

at a time

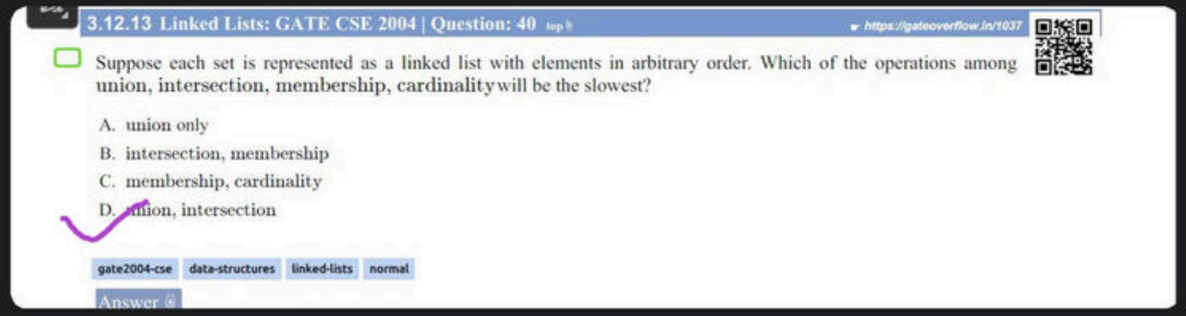
but both are implemented in system

no

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▲ 1 • Asked by Abhishek

Please help me with this doubt



3.12.13 Linked Lists: GATE CSE 2004 | Question: 40 top

https://gateoverflow.in/1037

Suppose each set is represented as a linked list with elements in arbitrary order. Which of the operations among union, intersection, membership, cardinality will be the slowest?

A. union only
B. intersection, membership
C. membership, cardinality
D. union, intersection

gate2004-cse | data-structures | linked-lists | normal

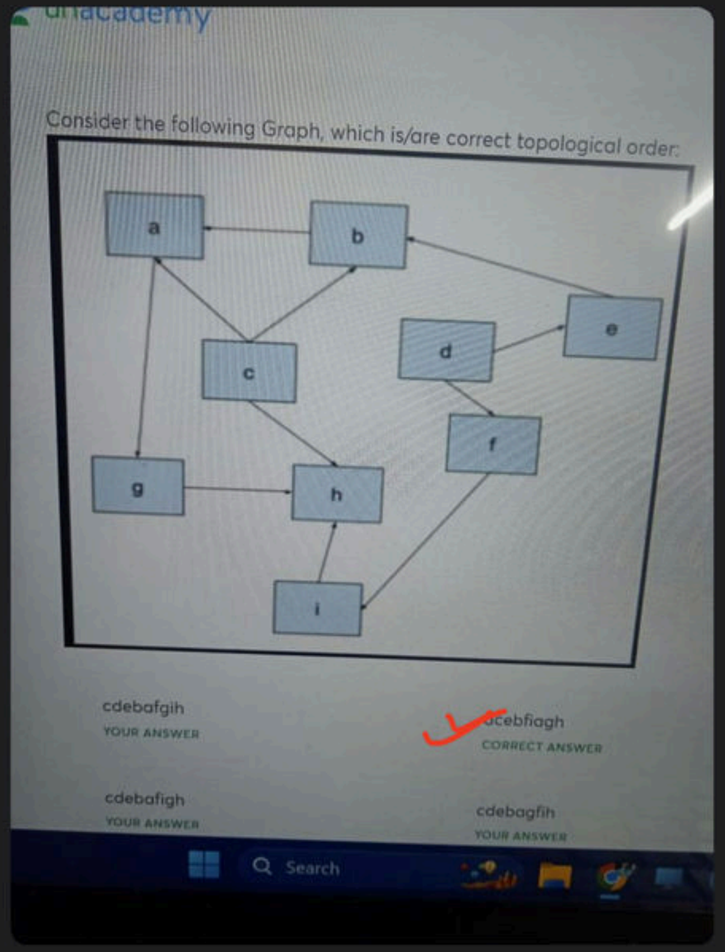
Answer

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▲ 1 • Asked by Abhishek

this is a DS test

que sir I have confusion on how the topological order is determined exactly



▲ 1 • Asked by Diksha

iska answer bta dijiye sir

Question 3 MSQ

Which of the following is/are true?

a) Any IO operation can be performed always after current instruction execution i CPU
b) DMA controller can start data transfer immediately after getting DMA request fror IO
c) CPU generates memory read/write signal for DMA transfer
d) CPU generates Interrupt acknowledgement immediately after receiving th interrupt always.

▲ 1 • Asked by Rishabh

sir in this Ques word length is also given then won't this make this Memory Word addressable? Then why have you solved considering it as Byte addressable?

